WORKED SOLUTION

STEP I 1988, Question 1

Pure • Curve Sketching

STEP I 1988, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q1

The key idea

The equation $n^m = m^n$ mixes the variables up between base and exponent, which is awkward. Taking logarithms gives $m \ln n = n \ln m$, and dividing by mn separates them completely: $\frac{\ln n}{n} = \frac{\ln m}{m}$. So we need two different integers at which h takes the same value, and the shape of the graph tells us exactly where such pairs can live.

Solution

The graph of h

Differentiating with the quotient rule,

$$\mathrm{h}'(x) = \frac{x \cdot \frac{1}{x} - \ln x}{x^2} = \frac{1 - \ln x}{x^2}.$$

Since $x^2 > 0$, the sign of $\mathrm{h}'(x)$ is the sign of $1 - \ln x$. So h is strictly increasing on $(0, \mathrm{e})$ and strictly decreasing on (e, ∞), with a maximum at $x = \mathrm{e}$ of value

$$\mathrm{h}(\mathrm{e}) = \frac{1}{\mathrm{e}}.$$

Differentiating again,

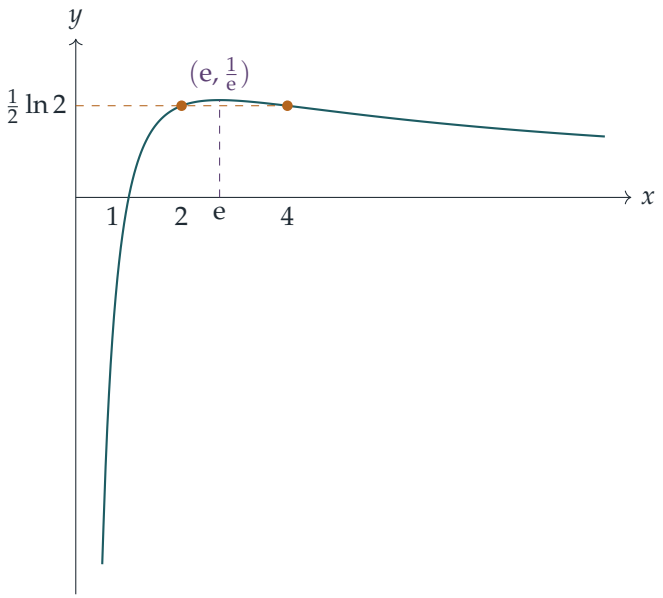
$$\mathrm{h}''(x) = \frac{-\frac{1}{x} \cdot x^2 - (1 - \ln x) \cdot 2x}{x^4} = \frac{2 \ln x - 3}{x^3},$$

so there is a point of inflection at $x = \mathrm{e}^{3/2} \approx 4.48$, where $\mathrm{h} = \frac{3}{2}\mathrm{e}^{-3/2} \approx 0.335$. The curve is concave to the left of this point and convex to the right.

Next, the behaviour at the ends and the sign:

- As $x \to 0^+$, $\ln x \to -\infty$ and $\frac{1}{x} \to +\infty$, so $\mathrm{h}(x) \to -\infty$: the y-axis is an asymptote.
- $\mathrm{h}(1) = 0$, and $\mathrm{h}(x)$ has the sign of $\ln x$: negative on $(0, 1)$, positive for $x > 1$.

- As $x \to \infty$, $\frac{\ln x}{x} \to 0$ (logarithms grow more slowly than any positive power of x), and $\mathrm{h}(x) > 0$ there, so the curve approaches the x-axis from above.



The equation $n^m = m^n$

Since m and n are positive, both sides are positive and we may take logarithms:

$$n^m = m^n \iff m \ln n = n \ln m \iff \frac{\ln n}{n} = \frac{\ln m}{m} \iff \mathrm{h}(n) = \mathrm{h}(m),$$

dividing by $mn > 0$ in the middle step. So we want distinct positive integers with $\mathrm{h}(m) = \mathrm{h}(n)$. By symmetry, suppose $m < n$.

Ask yourself Where on the graph can two different points have the same height?

Since h is strictly increasing on $(0, \mathrm{e}]$ and strictly decreasing on $[\mathrm{e}, \infty)$, it is one-to-one on each of these intervals. So two distinct inputs with the same output must lie on opposite sides of e: $m < \mathrm{e} < n$. Since m is a positive integer and $\mathrm{e} \approx 2.718$, $m = 1$ or $m = 2$.

- If $m = 1$ then $\mathrm{h}(n) = \mathrm{h}(1) = 0$, which forces $\ln n = 0$, so $n = 1 = m$, not allowed. (Directly: $n^1 = 1^n$ gives $n = 1$.)
- If $m = 2$ then we need an integer $n \geq 3$ with $\mathrm{h}(n) = \frac{1}{2}\ln 2$. Now $n = 4$ works, because

  $$\mathrm{h}(4) = \frac{\ln 4}{4} = \frac{2\ln 2}{4} = \frac{\ln 2}{2} = \mathrm{h}(2),$$

  that is, $4^2 = 16 = 2^4$. Since h is strictly decreasing on $[\mathrm{e}, \infty)$, it takes the value $\frac{1}{2}\ln 2$ at most once there, so $n = 4$ is the only possibility.

Hence the only pairs of distinct positive integers with $n^m = m^n$ are

$$(m, n) = (2, 4) \quad \text{and} \quad (m, n) = (4, 2).$$

Checks

A computer search over all $1 \leq m, n \leq 300$ with exact integer arithmetic finds only $(2, 4)$ and $(4, 2)$. Numerically, $\mathrm{h}(2) = \mathrm{h}(4) = 0.346574\ldots$, just below the maximum $1/\mathrm{e} = 0.367879\ldots$, as the graph suggests. You can also see why $\mathrm{h}(3)$ gives nothing: $\mathrm{h}(3) \approx 0.366$ exceeds $\mathrm{h}(2)$, and its partner on the left of e lies strictly between 2 and e, so it is not an integer.

Common slips

- Finding $(2, 4)$ by trial and stopping there. The question asks for *all* pairs, so the monotonicity argument proving there are no others is the heart of the answer.
- Forgetting the asymptotic behaviour in the sketch: the curve plunges to $-\infty$ at the y-axis but only creeps towards 0 as $x \to \infty$, never crossing the axis again.

Beyond STEP

If you allow real numbers, every a in $(1, \mathrm{e})$ has exactly one partner $b > \mathrm{e}$ with $a^b = b^a$, and these pairs can be parametrised neatly as $a = (1 + \frac{1}{t})^t$, $b = (1 + \frac{1}{t})^{t+1}$ for $t > 0$; as $t \to \infty$ both tend to e, which is the meeting point at the top of the graph. The limit $\frac{\ln x}{x} \to 0$ that we used is a first example of comparing growth rates, made rigorous in a first analysis course; see Abbott, *Understanding Analysis*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.